



The system Matra built and left — a VAL256 near Taipei Zoo, a decade after the walkout.

CHENG-EN CHENG [https://commons.wikimedia.org/wiki/File:Taipei_MRT_VAL_256_train_near_the_MRT_Taipei_Zoo_station_2007-07-11.jpg] • CC BY-SA 2.0 [<https://creativecommons.org/licenses/by-sa/2.0/>] • WIKIMEDIA COMMONS

The Matra dispute

Two disputes thirteen years apart, usually told as one — the test-track fires an engineering change explains, the seventeen-year lawsuit Matra won, and the walkout that taught Taipei how to buy its next railway.

Taiwan's first metro line was bought as a French turnkey system in 1988, caught fire twice on the test track in 1993, opened in 1996 more than four years late, and lost its builder's maintenance staff not long afterwards^[6]. Seventeen years after the signing, the Supreme Court ordered Taipei to pay Matra roughly **NT\$1.64 billion**, and the city paid^[1].

Almost all tellings run this together into one story about a French company storming off. It is two stories, thirteen years apart, and telling them apart is the point of this page.

BR WENHU LINE 文湖線			
CLIENT DORTS, Taipei City Government ^[1]	CONTRACTOR Matra Transport ^[1]	CONTRACT SIGNED 13 July 1988 ^[6]	CONTRACT VALUE ≈ NT\$7.87 billion ^[6]
LINE OPENED 28 March 1996 ^[9]	ARBITRATION AWARD 6 October 1993 ^[1]	FINAL JUDGMENT 22 July 2005 ^[1]	THE CITY PAID ≈ NT\$1.64 billion ^[1]

Two disputes, not one

The delay claim, 1988–2005.

Matra sued for the cost of a contract that ran years longer than it was written for, because the civil works its equipment depended on were late — the claim arises, in 自由時報's words, 「因土木工程部分包商施工延誤，連帶造成馬特拉的工期延長，增加成本費用」^[1]. This is the dispute that went to arbitration in 1993 and was still before the Supreme Court in 2005^[1]. It is about money and contract law, and **Matra won it**^[1].

The withdrawal, 1996–97.

After the line opened, Matra pulled its technical staff out of Taiwan and stopped maintaining the system^[7]. This is the dispute everybody remembers — the one that produced the most famous sentence said about the Taipei Metro `[/rail/operators/trtc/]`, and the one that steered the procurement that followed.

They are connected: the withdrawal happened because the delay claim had poisoned the relationship. But they are not the same event, and the NT\$1.64 billion that circulates belongs to the first, not the second^[1].

The contract

TERM	AGREED	SOURCE
Matra selected as preferred bidder	20 April 1988	[6]
Negotiations concluded	25 June 1988	[6]
Contract signed	13 July 1988	[6] [1]
Value	≈ NT\$7.87 billion	[6]
Scope	Muzha Line E&M, design, 51 two-car sets, testing, training	[6]

The scope matters more than the number. Taipei did not hire a contractor to electrify a railway; it bought a complete driverless system — vehicles, control, design and testing in one contract [1]. That single fact decides both halves of this story: it is why the courts eventually read the deal the way they did, and it is why losing the vendor meant losing the means to run the railway.

黃令名's account of the tender is that it was deliberately written to a 「歐美標準」 — a European-or-American standard rather than a US-only one — which opened it to European bidders, and that VAL was then the most mature automated system on the market [3]. He also records, without endorsing, that the award fed rumours connecting it to arms purchases from France [3]. He reports them as rumours; so does this page.

The fires, and the engineering behind them

Two fires, and what the builder calls them

There were two, four and a half months apart, and the second is recorded as exactly that: 「1993年5月5日：辛亥站前發生火燒車事件」 and 「1993年9月24日：發生第二次火燒車事件」 [5]. A second vehicle-fire incident. 自由時報 in 2005 said the same from the other end, describing 「接連被兩把大火燒得焦黑的電聯車」 — trains burned black by two successive fires [1].

That matters more than the count suggests. **One fire during testing is an accident. Two is a pattern**, and a pattern is what makes the rest of the year explicable: MOTC halting all work, the Control Yuan impeaching eight officials the following February, and a city council that then spent eight years refusing to let the same technology go north.

DORTS's own project record does not use the word fire at all. It says 「木柵線於82年系統測試階段發生輪胎起火意外事故」 — during the Muzha Line's system-testing phase in 1993, a **tyre ignition accident** [9]. One noun phrase, no date, no count, and it is the most precise wording anyone has used about the mechanism: what burned was a tyre, not a train.

Which of the two it refers to, the builder does not say. It may be either, or it may be the class of event rather than an instance of it. This page does not choose, because there is nothing to choose from.

So the two vocabularies are kept side by side rather than collapsed. The gap between 輪胎起火 and 火燒車 is roughly the gap between "an overheated tyre ignited" and "the train was on fire", which from a platform look identical — and the more precise phrasing belongs to the organisation that was impeached over it, which is worth seeing at the same time as the other.

The mechanism

There is an actual mechanism here, and it is better than "the French trains kept catching fire".

The trains were designed and run in France as **two-car units**. For Taipei the capacity specification became **four cars**, and the automatic train control was not redesigned to match — two two-car units were coupled and run on a shared signal path ^[3] ^[2]. When the link between them failed, the units disagreed about what to do next: in 黃令名's reconstruction the lead unit began to move while the trailing unit, having heard nothing, kept its brakes locked ^[3]; in 自由時報's, braking seized one axle pair while another drove on — 「煞車咬死，輪胎空轉摩擦過熱起火燃燒」 ^[2].

The two accounts differ in that detail and agree in everything that matters: a capacity change outran the signalling design, brakes dragged, and on rubber tyres against a concrete running surface, dragging makes heat and heat makes fire ^[3] ^[2]. On a steel-wheel railway the same failure produces wheel flats and noise. The fires were the tyres' version.

Two independent tellings, three years apart, is decent corroboration for a causal chain — but neither is an accident investigation, and none has been found. That gap is flagged again at the foot of this page.

The recorded incidents

- 5 May 1993 • **First fire**, during testing, 辛亥站前 — in front of **BR05** Xinhai ^[5]; attributed to 「煞車系統故障及人為疏忽」 — brake failure and human error; Matra apologised publicly on 14 May ^[6]
- From April 1993 • Cracked cap beams found at **BR07** Liuzhangli, **BR03** Wanfang Community and elsewhere; investigation and reinforcement demanded. ^[6]
- 19 June 1993 • Testing suspended after further smoke and fire events; in July Matra attributed them to 「晶片問題」 — chip faults. ^[6]
- 24 September 1993 • **Second fire**, recorded as 第二次火燒車事件, between **BR07** Liuzhangli and **BR06** Linguang. ^[5] ^[2]

- 1 July 1994 • Matra stops work, resuming on 23 July. [6]
- 25 August 1994 • Final inspections conclude. [6]
- 28 March 1996 • The line opens — four and a quarter years after the completion date the contract envisaged. [9] [6]

Most of this chronology is one encyclopedia entry deep [6]. The two fires themselves are better anchored: 自由時報 places them in May and September of 1993 [2], and its 2005 commentary invokes 「接連被兩把大火 燒得焦黑的電聯車，以及將近兩百根暴露著裂縫的捷運支柱」 — two burned trains and nearly two hundred cracked columns [1].

Those cracked columns deserve their own sentence. Cap beams are civil works, not Matra's electrical and mechanical scope — they belong to the other contractors on the project [1]. The same programme that burned Matra's trains in public was also, more quietly, running late underneath them, and that lateness is what Matra's claim was made of.

The arbitration, and seventeen years of litigation

From the 自由時報 report of the final judgment [1] — the best-documented part of the whole affair and the part least often told accurately:

- Matra's claim • 「廿多億元」 — over NT\$2 billion, for time added by others' delays [1]
- Negotiation • Collapsed [1]
- Arbitration, 6 October 1993 • For Matra: awards in NT dollars, US dollars and francs totalling ≈ **NT\$1.025 billion**, plus 5% annual interest [1]
- DORTS sues to annul the award • Loses; the Supreme Court upholds the award in November 2000 [1]
- DORTS resists enforcement • Argues the claim is time-barred: a works contract carries a short limitation period, expired since 1998. **Two courts agree.** [1]
- Supreme Court remands • The High Court, rehearing the case, reclassifies the contract as 「買賣、承攬的混合契約」 — a mixed sale-and-works contract — carrying a **fifteen-year** limitation. Matra is in time. [1]

Supreme
Court, 22
July 2005

DORTS's appeal dismissed. Principal plus twelve years' interest ≈
NT\$1.64 billion. ^[1]

Sit with the pivot for a moment, because it is the best irony in Taipei's transit history. The city very nearly won: two courts accepted that Matra had simply waited too long. What defeated it was the *character of the contract* — the rehearing found that a deal covering route definition, structural design, and the purchase, installation and testing of vehicles and equipment was not mere contracting but a purchase of a system, with a purchase's fifteen-year limitation ^[4]. **Taipei lost the lawsuit for the same reason it had lost control of its railway: it had bought a turnkey system.** The clause that cost it NT\$1.64 billion ^[1] and the lock-in that cost it its maintainer were the same decision, read twice.

The judgment itself has not been read here — only the reporting of it. The full sequence between 2000 and 2005 involves stages this page has deliberately left undated, because the accounts found disagree on the intermediate chronology and the judgment text would settle it. See the foot of the page.

The withdrawal, and the year nobody can pin down

That Matra pulled its people out is not in dispute. When, exactly, is.

〈木柵線〉 says it happened 「在通車兩個月之後」 — two months after the March 1996 opening, so around May 1996 ^[6]. The same encyclopedia's article on the through line says it happened in **1997**, after a contract dispute, when Matra 「將技術人員全數撤離，不再提供維修服務」 ^[7]. Both readings circulate; this site could not separate them, and no contemporaneous news report of the walkout itself has been found. One reconciliation — staff leaving through 1996, maintenance formally ending in 1997 — is plausible and is exactly the kind of guess this site does not publish as fact. **No year is asserted here.**

What the period looked like from inside city hall has, since March 2026, an account from the mayor himself: 陳水扁 describes a fight over spare-parts pricing — Matra quoting NT\$10,000 for a patented rubbish bin is his example — and the company using the threat of pulling its technicians as leverage in the renegotiation ^[4]. A memoir published on the line's thirtieth anniversary is a primary source with an interest in its own story, and it is cited here as that; but note that it frames the withdrawal not as a sudden exit but as a **negotiating instrument that was eventually used.**

商業周刊 was writing about the relationship under the title 「恩怨情仇」 — grudges and feuds — by 30 May 1996, six weeks after opening ^[8]. That article is the closest thing to a contemporaneous account of the collapse; it is paywalled, has not been read past its opening here, and is cited only for what its date and title establish: whatever the formal end date, the marriage was publicly over within weeks of the trains carrying passengers ^[8].

Keeping a metro nobody would maintain

What Taipei was left holding was a driverless system whose builder would not service it — proprietary control software, proprietary parts, and no second source for either.

黃令名's account — and this is his research speaking, singular and attributed, not a settled record — is that Taiwanese engineers kept the line running by 「偷偷複製法方半成品的列車控制程式、請台廠製造甚至赴光華商場取得 電子零件」: covertly copying the French side's half-finished train control software, commissioning parts from Taiwanese manufacturers, and sourcing electronics at Guanhua Market, Taipei's consumer electronics bazaar^[3]. He calls this 「技術能動性」 — technological agency — and it is fair to say his article exists to make that argument^[3].

陳水扁's retelling corroborates the substance from the client's side, minus the market stalls: replacement components were developed with 中研院, 中科院, 工研院 and private industry until every part had a domestic alternative, cheaper than the originals^[4]. Between a scholar's account of improvisation and a politician's account of mobilisation sits what actually happened; both agree the parts stopped coming from France and the trains kept running.

For the cost of the other road, 黃令名 points at Jacksonville, Florida — a city that faced a comparable dead end with its own Matra system and resolved it by demolition, rebuilding with a Bombardier monorail^[3]. Taipei's choice to keep the French railway alive without the French is the road that led, eventually, to this site's Wenhua Line [\[/rail/lines/wenhua-line/\]](/rail/lines/wenhua-line/) page describing one seamless operation.

The political ledger

「馬特拉不拉，我們自己拉」^[2] ^[4]

陳水扁, then Mayor of Taipei — *if Matra won't pull, we'll pull it ourselves*

The dispute consumed three mayors, one per act.

- 黃大洲 presided over the fires. 自由時報's commentary holds that the 木柵線 debacle 「間接導致了當時台北市長黃大洲的敗選下台」 — indirectly cost him the 1994 election^[1]. A newspaper asserting why an election was lost is a genre with a poor record; the timing fits, the claim is widely repeated, and it is published here as the paper's judgment rather than a fact.
- 陳水扁 opened the line on 28 March 1996^[6] and gave the affair the epitaph above. 自由時報 canonises it as his 名言^[2]; his own account of saying it, thirty years later, has it blurted out mid-crisis in the parts-pricing fight^[4]. The line is quoted everywhere and dated almost nowhere — no contemporaneous record of the moment of utterance has been found, and its setting rests on the speaker's memory.

- 馬英九 was mayor when the final bill arrived in July 2005, and said the city would pay what the judgment required^[1]. The DORTS director, 常岐德, declined all interviews and switched off his phone^[1]. The commentary's arithmetic was blunter: 「全民買單繳『學費』，足足要花十六億 四千萬元」 — the public pays the tuition, all NT\$1.64 billion of it^[1].

What it changed: buying the Neihu extension

When the line's northern half came to be tendered, DORTS changed the two things that had produced the trap, on 黃令名's analysis of the procurement^[3]:

1. **CBTC was specified** — communications-based train control, an open signalling approach — rather than an extension of Matra's proprietary fixed-block system^[3].
2. **Civil works and E&M were let together**, as one integrated contract, rather than split across contractors whose mutual delays had just spent seventeen years in court^[3].

By then the incumbent was not even Matra. Siemens had taken control of Matra Transport International in the 1990s — 95% by October 1998^[10] — and completed the takeover in November 2001, renaming the business Siemens Transportation Systems^[11]. VAL was now a Siemens product, and a tender written around VAL would have had exactly one credible bidder. 黃令名's argument is that the CBTC-plus-turnkey structure existed precisely so that Siemens could not name its price as the only party able to extend the existing system^[3]. Procurement rationales are the kind of thing institutions tidy up afterwards, and his is the only account found^[3] — but it is consistent with what was actually bought.

What was bought was Bombardier. And here the engineering gets genuinely elegant, because Bombardier could not supply a VAL — VAL belonged to a competitor. So it built the Innovia APM 256 [\[/rail/rolling-stock/innovia-apm-256-c370/\]](/rail/rolling-stock/innovia-apm-256-c370/): a people mover dimensioned to a French railway it had no rights to, matching the running-surface spacing, the side guide bars and the 750 V supply taken off them, with a car body 20 mm narrower than the VAL256 [\[/rail/rolling-stock/val256/\]](/rail/rolling-stock/val256/) it would run beside^[13]. The incumbent's guideway became the tender's interface specification.

The last act was the cutover. Rather than run two signalling systems, the whole line switched to Bombardier's CITYFLO 650 on the day the extension opened — **4 July 2009** — and the 1996 Matra fleet was withdrawn that same day for conversion, returning on 26 December 2010 after eighteen months with its French control system replaced^[12] ^[7]. 黃令名 adds — his account alone; nothing else found says it — that CITYFLO had been tested at night on the Muzha Line's own tracks while the Matra system carried passengers by day^[3]. It is a striking detail, it is single-source, and those two properties are related; it is recorded here with its attribution showing.

The last Matra-controlled train ran thirteen years after Matra stopped maintaining any of them. Nobody from the company was there to see it, and by then there was no company: the name had been gone from the rail business for eight years^[11].

What is not established

The gaps, recorded so they are searched for rather than papered over:

- **The contract itself.** No tender documents, competing bids or evaluation records were found; they would sit in DORTS's construction archive.
- **The judgments.** Taiwan's court decisions are indexed and searchable, and the 22 July 2005 Supreme Court civil judgment ought to be retrievable; it would move the whole litigation section from reportage to primary record, and settle the intermediate dates this page declines to give. The single highest-value follow-up here.
- **A contemporaneous report of the withdrawal,** which would close the question above of which year it happened in ^[6] ^[7].
- **Any accident investigation** into the 1993 fires, which would confirm or kill the coupling explanation two secondary sources agree on. DORTS's own classification of the testing-phase incident — 輪胎起火意外事故, a tyre ignition accident ^[9] — is as close as the primary record gets, and it names no date, no unit and no location.
- **Which fire DORTS's 輪胎起火意外事故 describes.** The builder names an incident and gives it no date and no ordinal ^[9]. There were two. Nothing found says which, or whether the phrase names the class rather than an instance.
- **Matra's side.** Every substantive source on this page is Taiwanese. The French trade press of the day, Lagardère's archive and Siemens's own history of the VAL business would give the other account, and none has been found. Its absence is the largest structural bias in this page, and it is not a small one.

References

3 of 13 sources on this page are primary — published by the operator, the builder or government.

- 1 Muzha metro line: Matra wins, Taipei to pay NT\$1.64 billion [<https://www.epochtimes.com/b5/5/7/23/n995027.htm>]

木柵捷運線 馬特拉勝訴 北市要賠16.4億

SECONDARY 孫友廉 , 陳曉宜 , 鄭學庸 , 自由時報 , 23 July 2005 (via 大紀元 reprint)
accessed 2026-08-06

archived [<https://web.archive.org/web/20260807043739/https://www.epochtimes.com/b5/5/7/23/n995027.htm>]

The contemporaneous report of the final judgment, with the figures, the procedural stages and the limitation-period reasoning, plus a signed commentary. The best source this page has for the litigation. The original 自由時報 URL has rotted; the 大紀元 reprint is what remains reachable.

- 2 Cars altered, trains burned — and Matra paid NT\$1.6 billion anyway [<https://news.ltn.com.tw/news/life/paper/258078>]

改車廂火燒車 還賠馬特拉16億

SECONDARY 洪敏隆 , 劉榮 , 自由時報 , 14 November 2008 accessed 2026-08-06

archived [https://web.archive.org/web/20260807043814/https://news.ltn.com.tw/news/life/paper/258078]

Found while preparing this page, and load-bearing: an account of the fire mechanism — two-car units run as four in Taipei, brakes locking on one axle pair while another drove on, tyres overheating — that is independent of and three years earlier than 黃令名's. Two tellings, one mechanism.

-
- 3 How the Taipei Metro became a 'jury-rigged metro': technological hybridity and the birth of the system [https://twstreetcorner.org/2021/04/06/台北捷運如何成為「拼裝捷運」？科技混生性與北捷/]

台北捷運如何成為「拼裝捷運」？科技混生性與北捷的誕生

SECONDARY 黃令名 , 巷仔口社會學 (Streetcorner Sociology), 6 April 2021

accessed 2026-08-06

archived [https://web.archive.org/web/20260807043954/https://twstreetcorner.org/2021/04/06/%E5%8F%B0%E5%8C%97%E6%8D%B7%E9%81%8B%E5%A6%82%E4%BD%95%E6%88%90%E7%82%BA%E3%80%8C%E6%8B%BC%E8%A3%9D%E6%8D%B7%E9%81%8B%E3%80%8D%E5%BC%9F%E7%A7%91%E6%8A%80%E6%B7%B7%E7%94%9F%E6%80%A7%E8%88%87%E5%8C%97/]

A sociologist of technology writing from his own doctoral research — still the most valuable single account of these events in any language. Not neutral, and does not claim to be: the framing is a thesis about technological dependence. Also republished by The News Lens.

-
- 4 Without Matra, we pulled it ourselves — thirty years of the Taipei Metro [https://www.mirrormedia.mg/story/20260305cu1002]

名家現場／沒有馬特拉，我們自己拉——北捷三十年

PRIMARY 陳水扁 , 鏡週刊 , 11 March 2026 accessed 2026-08-06

archived [https://web.archive.org/web/20260806043259/https://www.mirrormedia.mg/story/20260305cu1002]

The man who said it, on the record thirty years later. Primary for the quotation and for his own account of the parts-pricing fight behind it — and read with the caution any memoir earns: it is the speaker's version, published on an anniversary.

-
- 5 Linguang station [https://zh.wikipedia.org/zh-tw/麟光站]

麟光站

SECONDARY 維基百科 (Chinese Wikipedia) accessed 2026-08-07

The station chronology that dates both 1993 fires and, decisively, numbers the second:

「1993年5月5日：辛亥站前發生火燒車事件」 and 「1993年9月24日：發生第二次火燒車事件」. 〈木柵線〉 carries the May incident but not the September one, which is why an earlier pass on this page could not confirm the second date.

-
- 6 Muzha Line [https://zh.wikipedia.org/zh-tw/木柵線]

木柵線

SECONDARY 維基百科 (Chinese Wikipedia) accessed 2026-08-06

archived [https://web.archive.org/web/20250531163800/https://zh.wikipedia.org/zh-tw/木柵線]

The fullest chronology of the construction years — contract dates, the fires, the suspensions, the inspections. It also contradicts itself on the year Matra withdrew, which this page publishes as the disagreement it is.

-
- 7 Wenhua line [https://zh.wikipedia.org/zh-tw/文湖線]

文湖線

SECONDARY 維基百科 (Chinese Wikipedia) accessed 2026-08-06

archived [https://web.archive.org/web/20250714045049/https://zh.wikipedia.org/zh-tw/文湖線]

Carries the 1997 reading of the withdrawal — 馬特拉與台北市政府捷運工程局發生合約糾紛後，將技術人員全數撤離。

- 8 The Taipei Metro's Matra feud [https://www.businessweekly.com.tw/Archive/Article?StrId=1967]

台北捷運的「馬特拉」恩怨情仇

SECONDARY 古文全，商業周刊 445, 30 May 1996 accessed 2026-08-06

archived [https://web.archive.org/web/20260807044117/https://www.businessweekly.com.tw/Archive/Article?StrId=1967]

Written six weeks after the line opened — the closest thing to a contemporaneous account of the relationship breaking down. Paywalled beyond its opening, and not read past it here; cited only for what its date and title establish.

- 9 Wenshan—Neihu Line [https://www.dorts.gov.taipei/cp.aspx?n=DBAC040496EFAB94]
文山內湖線

PRIMARY Taipei City Government, Department of Rapid Transit Systems (DORTS)

accessed 2026-08-07

archived [https://web.archive.org/web/20260209210832/https://www.dorts.gov.taipei/cp.aspx?n=DBAC040496EFAB94]

The builder's own project record — section lengths, opening dates, and its own classification of the testing-phase incident: 「木柵線於82年系統測試階段發生輪胎起火意外事故」。It says nothing at all about the dispute, which is itself informative; and it names Matra nowhere.

- 10 Siemens increases its stake in MATRA Transport International [https://www.lagardere.com/en/press-release/siemens-increases-its-stake-in-matra-transport-international/]

PRIMARY Lagardère (press release, 1 October 1998) accessed 2026-08-06

archived [https://web.archive.org/web/20260806042121/https://www.lagardere.com/en/press-release/siemens-increases-its-stake-in-matra-transport-international/]

Matra's own parent announcing the sale of the transport business to Siemens.

- 11 Matra Transport changes its name [https://www.metro-magazine.com/news/matra-transport-changes-its-name]

SECONDARY Metro Magazine accessed 2026-08-06

archived [https://web.archive.org/web/20260807044158/https://www.metro-magazine.com/news/matra-transport-changes-its-name]

Trade press reporting Siemens completing to 100% and the rename to Siemens Transportation Systems in November 2001.

- 12 Taipei Metro VAL256 electric multiple unit [https://zh.wikipedia.org/zh-tw/台北捷運VAL256型電聯車]

台北捷運VAL256型電聯車

SECONDARY 維基百科 (Chinese Wikipedia) accessed 2026-08-06

archived [https://web.archive.org/web/20221006051618/https://zh.wikipedia.org/zh-tw/台北捷運VAL256型電聯車]

13 **Bombardier INNOVIA APM 256 electric multiple unit** [https://zh.wikipedia.org/zh-tw/龐巴迪INNOVIA_APM_256型電聯車]
龐巴迪INNOVIA APM 256型電聯車
SECONDARY 維基百科 (Chinese Wikipedia) accessed 2026-08-06
archived [https://web.archive.org/web/20150531215555/https://zh.wikipedia.org/zh-tw/龐巴迪INNOVIA_APM_256型電聯車]

Last updated: 2026-08-06

An independent, non-commercial reference site. Not affiliated with Taipei Rapid Transit Corporation, Taipei City Government, or any bus operator. Line colours are the official values published by each operator through Taiwan MOTC's open data platform. [About this site.](#)

Station names, codes and sequence from TDX Transport Data eXchange, Ministry of Transportation and Communications, operator TRTC. Government open data. Retrieved 2026-08-10; source last updated 2020-01-31. Provenance.